
Quantum Computing

*Laurea Magistrale in Ingegneria e Scienze
Informatiche*

AA 2025-2026



UNIVERSITÀ
di **VERONA**

Dipartimento
di **INFORMATICA**

About the Course

- Introducing the basic concepts of Quantum Computing via the study of the physical phenomena characterising this paradigm with respect to the classical one, and the algorithmic approaches exploiting these phenomena.
- Lectures on
 - Tuesday's (14:00 - 15:30) in Room B
 - Friday's (14:30 - 16:00) in Room L
- References and teaching material available on [Moodle](#).

Other resources:

- Quantum Programming environments with great documentation: [Qiskit](#), [Q#](#), [Cirq](#), [QPS](#), [BlueQubit](#) platform
- Lectures by [Scott Aaronson](#), Scott Aaronson's blog, [David Deutsch](#) and Ronald de Wolf
- Lecture on Quantum Computer Programming by Will Zeng

Testi Consigliati

Libri consigliati

[Vai a sezione](#)

[An introduction to quantum computing / Phillip Kaye, Raymond Laflamme, Michele Mosca](#)

Autore: Kaye, Phillip **Altro Autore:** Laflamme, Raymond; Mosca, Michele **Tipo:** Libro **ISBN:** 0198570007

Numero OCLC: 297561 **Editore:** Oxford University Press **Luogo di pubblicazione:** Oxford **Data di**

pubblicazione: 2007 **Disponibile a:** Biblioteca "Bruno Forte", Dipartimento di Informatica Consultazione INF MAS 015; Biblioteca "Bruno Forte", Dipartimento di Informatica Prestito 20 giorni INF 511 KAY; Biblioteca "Bruno Forte", Dipartimento di Informatica Prestito 20 giorni INF 511 KAY/A

[Vai a copia](#)

[Quantum computation and quantum information / Michael A. Nielsen and Isaac L. Chuang](#)

Autore: Nielsen, Michael A. **Altro Autore:** Chuang, Isaac L. **Tipo:** Libro **ISBN:** 0521632358 **Numero OCLC:**

273884 **Editore:** Cambridge University Press **Luogo di pubblicazione:** Cambridge **Data di pubblicazione:** c2000 **Disponibile a:** Biblioteca "Bruno Forte", Dipartimento di Informatica Prestito 20 giorni INF 511 NIE; Biblioteca "Bruno Forte", Dipartimento di Informatica Prestito 20 giorni INF MAN 033

[Vai a copia](#)

Testi Consigliati

[Programming quantum computers : essential algorithms and code samples](#)

Autore: Johnston, Eric R., author. **Altro Autore:** Harrigan, Nic, author.; Gimeno-Segovia, Mercedes, author.

Tipo: Libro **ISBN:** 1492039632 **Numero OCLC:** (ckb)4100000008693759 **Editore:** O'Reilly **Luogo di**

pubblicazione: Beijing **Data di pubblicazione:** 2019 **Edizione:** First edition. **Note:** Includes bibliographical references and index.

[Vai a copia](#)

[Quantum computing : an applied approach](#)

Autore: Hidary, Jack D., author. **Tipo:** Libro **ISBN:** 978-3-030-83273-5 **Numero OCLC:** 1272992416 **Editore:**

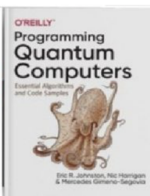
Springer, **Luogo di pubblicazione:** Cham, Switzerland : **Data di pubblicazione:** 2021 **Edizione:** Second edition.

DOI: <https://doi.org/10.1007/978-3-030-83274-2>

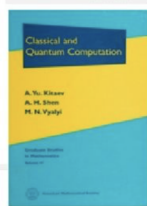
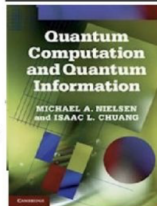
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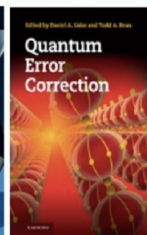
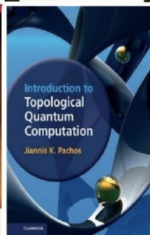
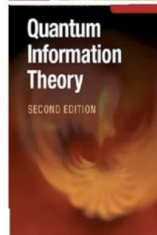
Basics ➡



Intermediate ⇨



Advance ⇨



Background

- Physics
No knowledge of physics is required!
- Complex Numbers
- Linear Algebra
Concepts of basis, dimension, norm, inner product, orthogonality, matrices, adjoint, and eigenvalues.
- Probability
- Programming (for the lab).

A Century of Quantum



INTERNATIONAL YEAR OF Quantum Science and Technology

On June 7, 2024, the United Nations proclaimed 2025 as the International Year of Quantum Science and Technology (IYQ). The 2025 IYQ recognizes 100 years since the initial development of quantum mechanics.

The Birth of Quantum Mechanics

- Max Planck (1900): Energy is emitted and absorbed in discrete **quanta**. **Nobel 1918**
- Albert Einstein (1905): Light as Particles - Photoelectric effect. Light is quantized in **photons**. **Nobel 1921**
- Niels Bohr (1913): Model of the atom - **Quantised Orbits**. **Nobel 1922**
- Louis de Broglie (1924): The wave nature of particles. **Nobel 1929**
- Werner Heisenberg (1925): Matrix mechanics - Quantum states and observables as matrices **Nobel 1932**
- Schrödinger (1925): Wave mechanics - Quantum states described by a wave function **Nobel 1933**
- Max Born (1926): Statistical interpretation of the wave function **Nobel 1954**

The 1927 Solvay Conference on Physics



The 5th Solvay Conference on Physics



The conference marked the split between Einstein and Bohr on the interpretation of Quantum Mechanics. The Copenhagen Interpretation was absurd also for Schrödinger, de Broglie and others who had created Quantum Mechanics.

Interpretations of Quantum Mechanics

Quantum Theory and Einstein's **General Theory of Relativity** are the two great fundamental theories of contemporary Physics. They provide

- the conceptual framework and the mathematical language in which we express all other theories in Physics, and
- the basic principles to which all known laws of Nature conform.

The deeper and more general a theory is, the more it tends to involve unfamiliar phenomena. Although the outcomes of quantum mechanical experiments are all uncontroversial, the underlying explanation is still very controversial.

- Many rival explanations or **Interpetations of Quantum Theory**.
- The **Copenhagen interpretation** is the oldest and, collectively, probably still the most widely held attitude about quantum mechanics.

The Copenhagen Interpretation

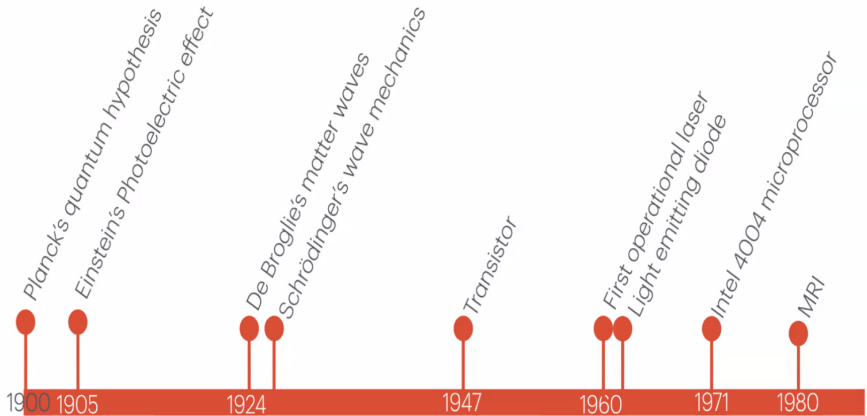
- Emerged from Niels Bohr's Institute in Copenhagen
- Key idea: the act of observation or measurement, which induces the collapse of the state (wave function) selecting a definite outcome from a probabilistic description.
- Measurement problem: the theory does not tell where the threshold between the microscopic and macroscopic worlds is, that is when quantum mechanics should leave space to classical mechanics.

Other theories have been introduced to overcome the **measurement problem**, e.g. **Hugh Everett's many-worlds interpretation**, **Bohmian Mechanics**, **Retro-causality** (or transactional interpretation), **Quantum Logic**.

Watch this **video** for a nice explanation of the different interpretations.

The First Quantum Revolution

Quantum Mechanics becomes technology: From quantum theory to laser, transistors etc.



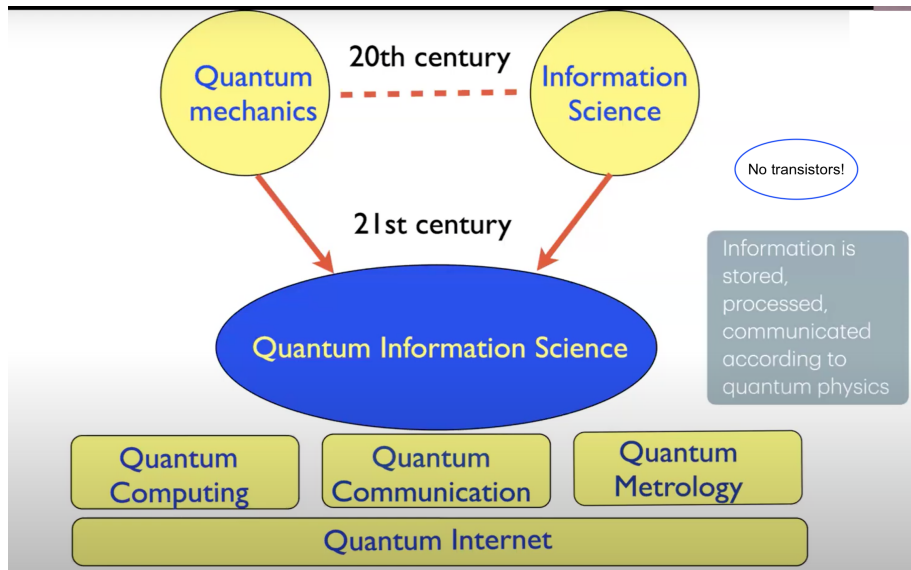
The Second Quantum Revolution

The ability to control a single quantum system (e.g. ions or photons) was developed only after the 1980's.

- Hans Dehmet (1986): Ion-trap technique. Nobel 1989
- S. Haroche and D.J. Wineland: Experimental methods that enable measuring and manipulation of individual quantum systems. Nobel 2012
- A. Aspect, J.F. Clauser and A. Zeilinger: experiments with entangled photons, establishing the violation of Bell inequalities. Nobel 2022

The latter result allowed to transform the "Spooky Action" into Quantum Technology, marking the birth of Quantum Information Science.

The Second Quantum Revolution



The Second Quantum Revolution

Key technologies for Industry 4.0

Quantum
Metrology

Data collection by precision measurements

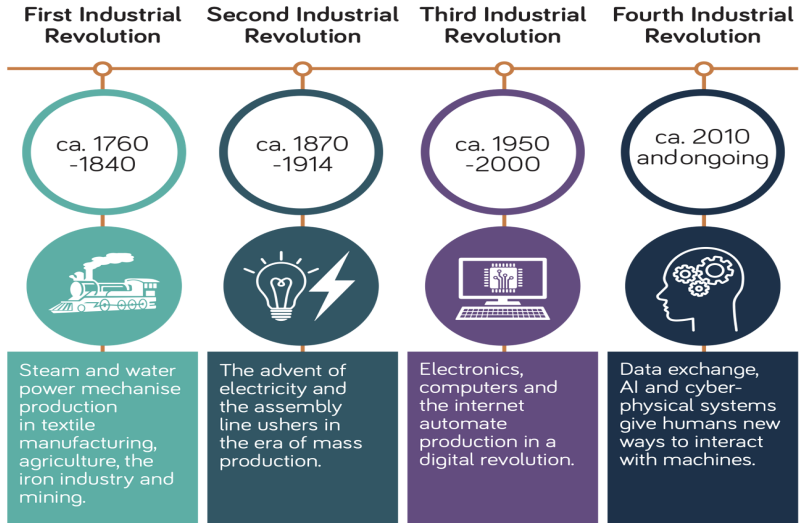
Quantum
Communication

Safe **Data** transportation

Quantum
Computing

Ultra-fast **Data** Processing

The Industry Revolutions



Quantum Computing

Today's classical computers don't count as quantum computers even though their microchips rely on quantum mechanics (they are very small).

- The repertoire of computation is still the same as the abstract universal Turing Machine.
- Quantum computers are capable of new modes of computation which are unavailable on classical computers even in principle.

In a multiverse interpretation,

- Quantum computation is the full theory which has the multiverse as the arena.
- Classical computation is the approximation occurring in self-contained universes, which don't affect each other.

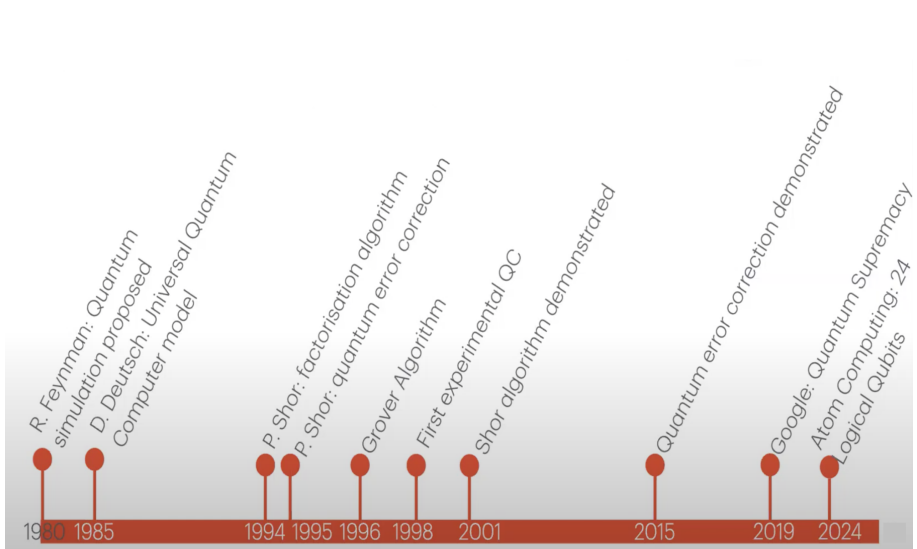
Physics and Computation

The theory of computation was originally conceived of as a branch of pure mathematics. We can now say that it always did have a lot in common conceptually with physics.

Computation		Physics
Computer		Physical System
Computation		Motion
Input		Initial state
Rules		Laws of motion
Output		Final state

Quantum computation is the reformulation of the theory of computation in Quantum Physics.

A Timeline for Quantum Computing



Quantum Speedup

Description	Classical Algorithm	Quantum Algorithm	Reference
Grover's algorithm: ✓ searches an unstructured database (or an unordered list) with N entries	$O(N)$	$O(\sqrt{N})$	L.K. Grover, "A fast quantum mechanical algorithm for database search." Proceedings of the twenty-eighth annual ACM symposium on Theory of computing. ACM(1996).
Shor's algorithm: ✓ Integer factorization	$O\left(e^{1.9(\log N)^{\frac{1}{3}}(\log \log N)^{\frac{2}{3}}}\right)$	$O\left((\log N)^2(\log \log N)(\log \log \log N)\right)$	D. Beekman, et al. "Efficient networks for quantum factoring." Physical Review A 54. 2 (1996).
Quantum Fourier Transform: ✓ Discrete Fourier transform of size N	$O(N \log N)$	$O(\log N \log \log N)$	P. Shor, "Algorithms for quantum computation: discrete logarithms and factoring," Proc. 35th Annual Symp. on Foundations of Comp. Sci., pp.124-134(1994)
Eigenvalue solver: ✓ To find eigenvalues and eigenvectors of a local Hamiltonian	$O(N^{2 \sim 2.236})$	$O\left((\log N)^4\right)$	Abrams and S. Lloyd, "Quantum Algorithm Providing Exponential Speed Increase for Finding Eigenvalues and Eigenvectors," Phys. Rev. Lett. 83, 24, p.5162 (1999).
Matrix inversion: ✓ Finding inverse matrix. This can be applied to find a vector x satisfying $Ax=b$, where A and b are Hermitian $N \times N$ matrix and a unit vector, respectively	$O(Nsk(\log 1/\epsilon))$	$\tilde{O}\left(\frac{\log N s^2 \kappa^2}{\epsilon}\right)$	A.W. Harrow, Avinandan Hassidim, and S. Lloyd. "Quantum algorithm for linear systems of equations." Physical review letters 103.15 (2009): 150502.
Distance (inner product) evaluation: ✓ Calculating inner product between a given N dimensional vector and each N dimensional vector of M samples	$O(MN)$	$O(\log NM)$	S. Lloyd, Masoud Mohseni, and Patrick Rebentrost. "Quantum algorithms for supervised and unsupervised machine learning." arXiv preprint arXiv:1307.0411 (2013).

Computability

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- The answer may come from a theory reconciling **Einstein's general theory of relativity** and the **standard model of particle physics**.

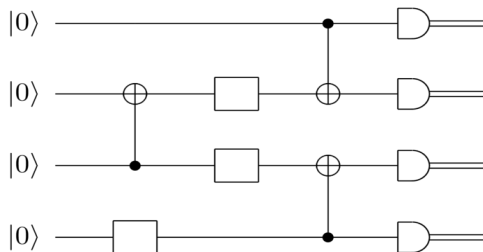
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- The answer may come from a theory reconciling **Einstein's general theory of relativity** and the **standard model of particle physics**.
- Theory of quantum gravity ?

The Circuit Model

General form of a quantum circuit:



- Start in a computational basis state.
- Apply a sequence of CNOT and single-qubit gates.
- To obtain the result, measure in the computational basis.

Quantum Mechanics

See Lectures on 'Introduzione alla meccanica quantistica per il quantum computing' (2025/2026, prof.ssa Daffara)